

Acetaminophen Project

By Kaiser Lee and Vivian Pellock

User's Manual

The Acetaminophen metabolism simulation is a deterministic system dynamics model that generates multiple plots showing the amount of chemicals at different parts of the body involved in the metabolization of acetaminophen.

To run the simulation, install the project files. Then open `Ace_Main.py` and run it.

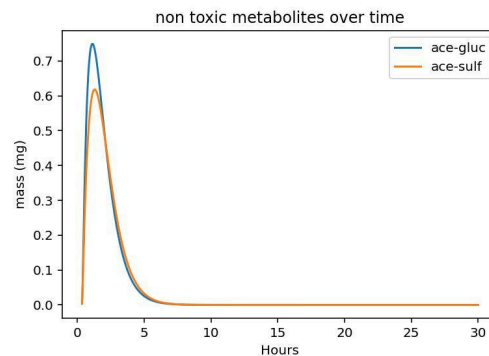
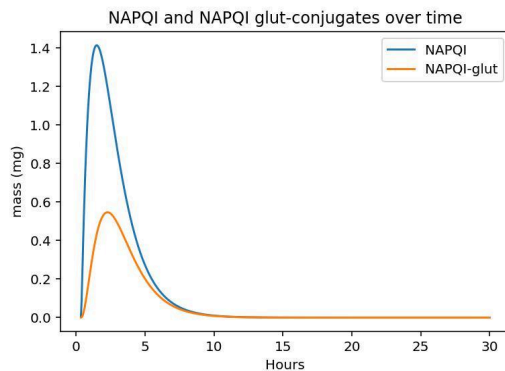
To adjust the variables of the simulation, go into the `ace_global_vars.py` file and change the variables to simulate under different variables. These are the main variables intended to be adjusted for the simulation:

- `fig_title_prefix` is a string which labels the image files of the plots generated.
- `fig_dir` variable determines the directory the plots are generated in.
- `save_plt` is a bool variable that determines if the plots you generate with the simulation will be saved as images. If you wish to generate plots, set `save_plt` as `True`, otherwise, set it to `False`.
- `time` variable determines the amount of time the simulation will simulate in hours. For example, if the time is set to 24, then it will simulate the metabolization of acetaminophen over 24 hours after first dose.
- `dose_interval` determines the time between doses if there are multiple doses in the simulation.
- `dose_count` determines the amount of doses will be taken in the simulation. If it is set to 1, it will simulate only one dose. If it is set to value more than 1, then it will dose multiple times.
- `dose_amount` determines the amount of acetaminophen taken per each dose.
- `ethanol_amount` determines the amount of alcohol in the system at the start of simulation.
- Rest of the variables determine the other constants used for the calculations within the simulation.

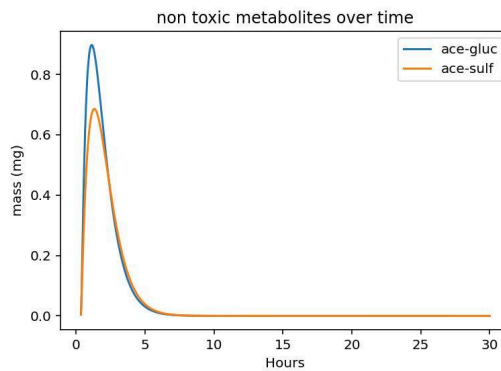
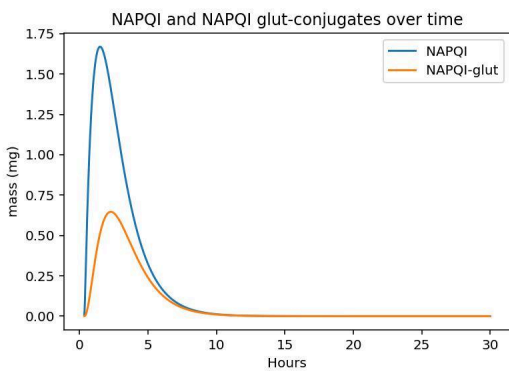
Analysis

What effect does one large dose (over 1.5 grams) have on the body and enzyme reserves?

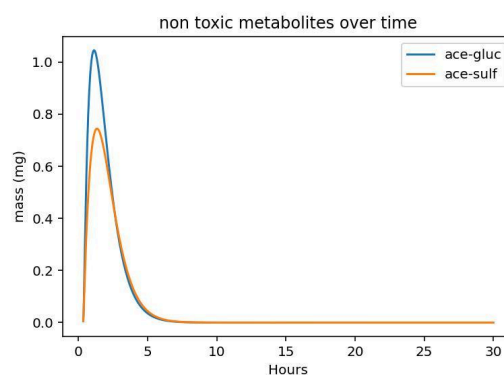
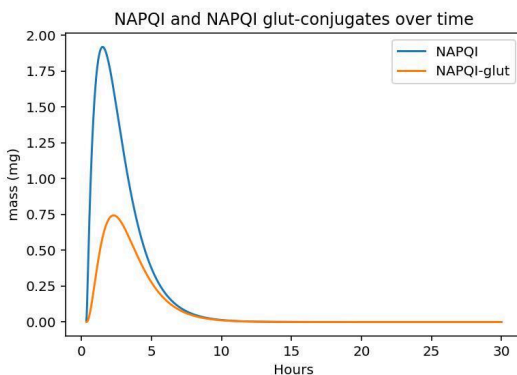
Single 1250 mg dose



Single 1500 mg dose



Single 1750 mg dose



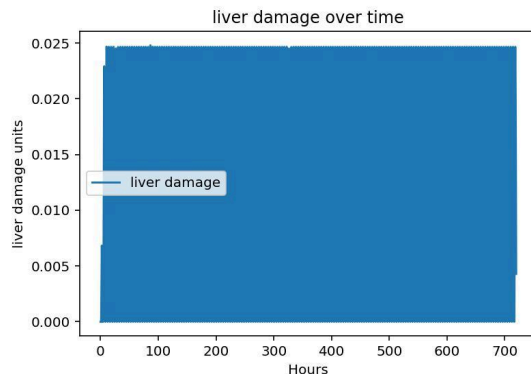
Comparing the result of a dose under 1500 mg, 1500 mg, and dose over 1500 mg, we found that the body seems to be able to metabolize acetaminophen successfully, but when the dose is higher, all metabolites and NAPQI conjugate reaches a higher peak.

What effect does longer-term use of proper doses have on the body and enzyme reserves?

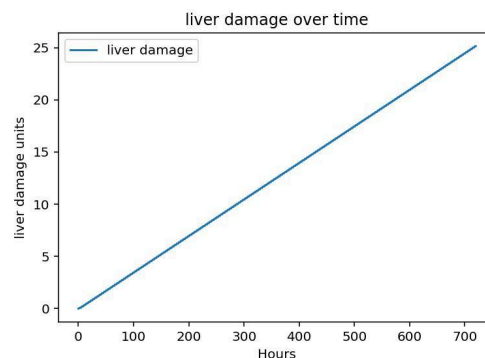
According to MayoClinic.org,

(<https://www.mayoclinic.org/drugs-supplements/acetaminophen-oral-route-rectal-route/description/drg-20068480>) the recommended dose of acetaminophen for oral dosage form is 650 to 1000 mg every 4 to 6 hours for adults. The following are the result of dosing 650mg every 4 hours, 650mg every 6 hours, 1000mg every 4 hours, and 1000mg every 6 hours for 30 days (720 hrs):

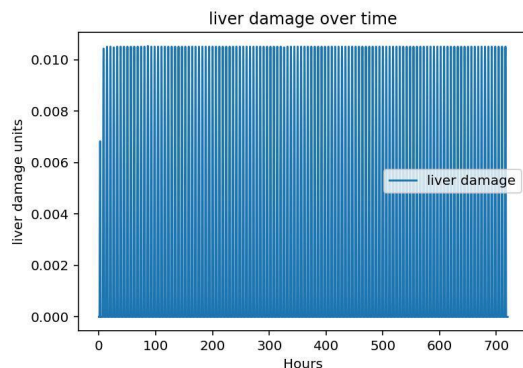
650 mg every 4 hrs - liver damage



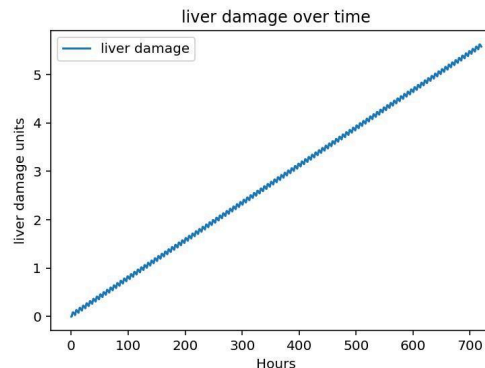
1000 mg every 4 hrs - liver damage



650 mg every 6 hrs - liver damage



1000 mg every 6 hrs - liver damage



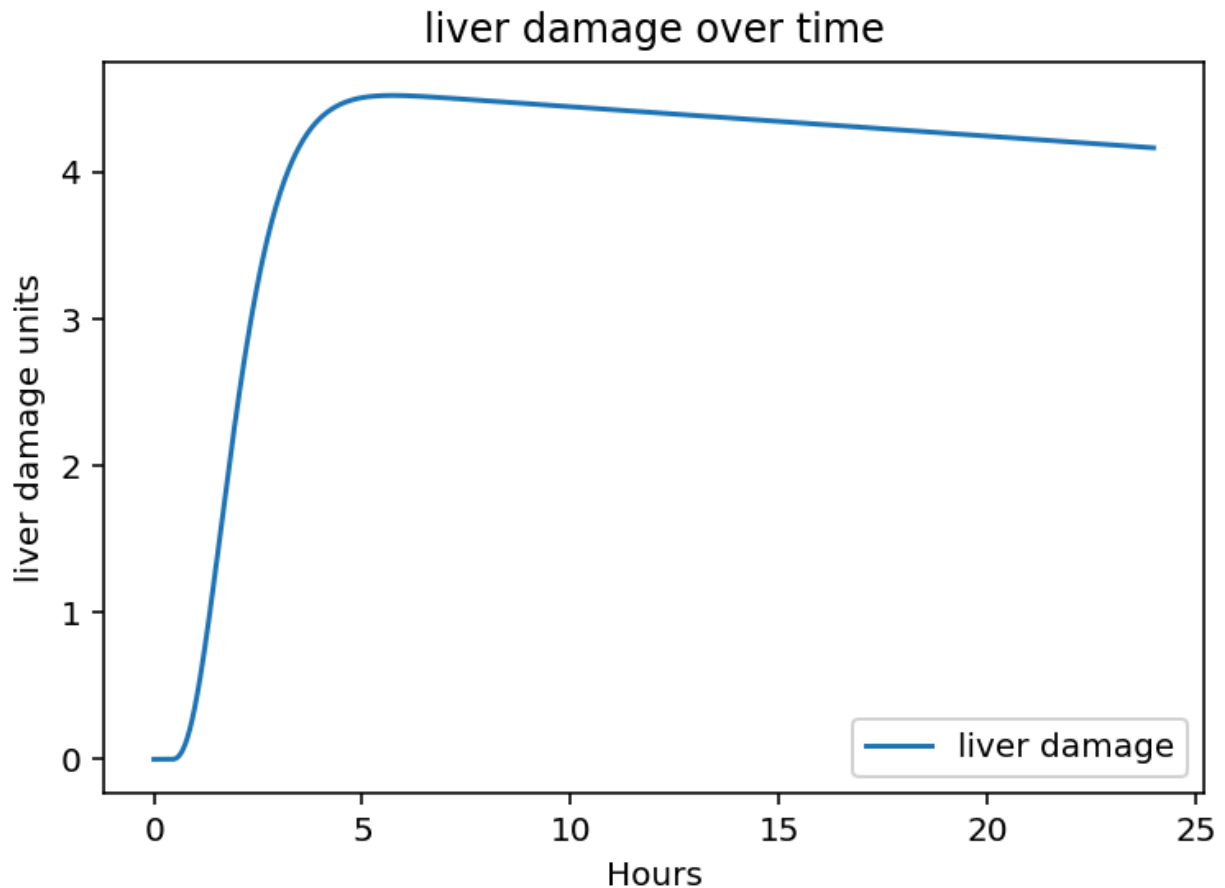
We expect that cases where less than 3000 mg of acetaminophen was taken per day would result in no damage while taking over 3000 mg per day would result in damage.

In our simulation, liver damage units of 8 is when hepatocyte viability goes to 0, which is considered fatal. Based on the simulation results, taking 650 mg per 4 hours and per 6 hours is both safe. However, taking 1000 mg every 6 hrs is dangerous as it results in damage units over 5, and taking 1000 mg every 4 hrs is fatal.

From this result, we concluded that taking about 1000 mg per dose is dangerous to do over a long period of time, but 650 mg appears to be safe enough for long term use.

How far into overdose does one have to go for the damage to be considerable?

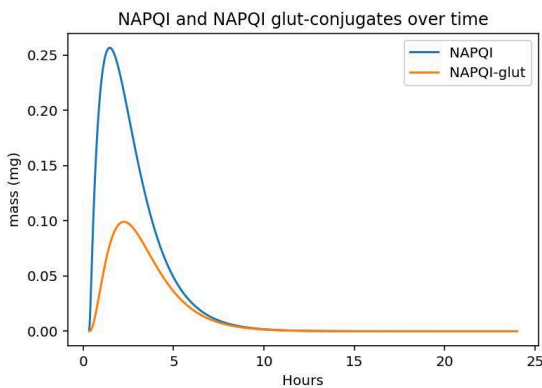
We will define considerable damage to rank at a 4 on the scale, considering that a slight (but likely not very harmful) overdose is about a 1.3. Using these assumptions, then you would have to take 5.5 grams of acetaminophen in order to have considerable damage from it.



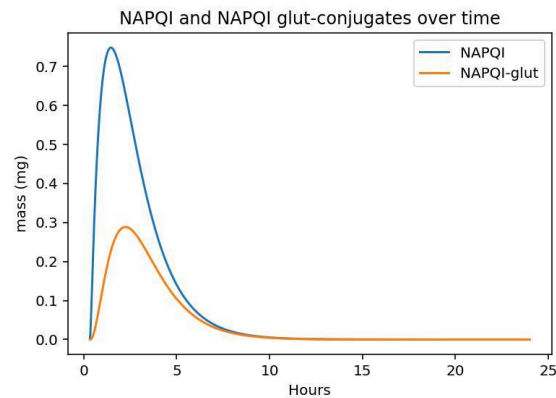
What effect does alcohol consumption have on NAPQI levels and metabolism?

One 12 fl oz can of beer contains about 305 mmol of ethanol. We hypothesize that consumption of alcohol will result in increased NAPQI levels. The following are the result of having had no alcohol, 1 can of beer (305 mmol) and 6 cans of beer (1830 mmol) while taking 1 dose of 650 mg of acetaminophen.

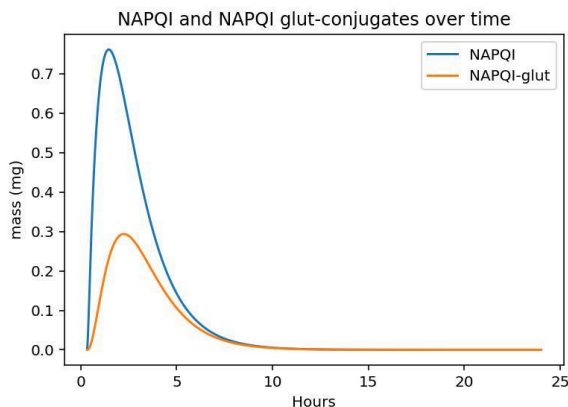
0 mmol of ethanol



305 mmol of ethanol



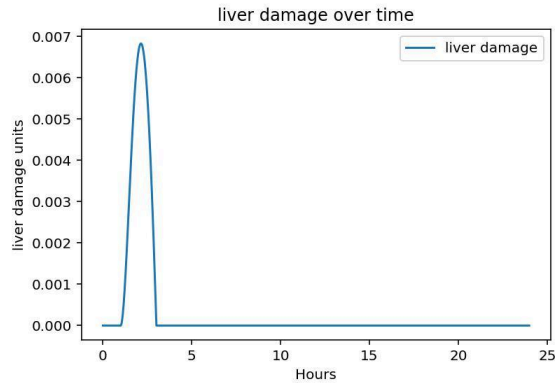
1830 mmol of ethanol



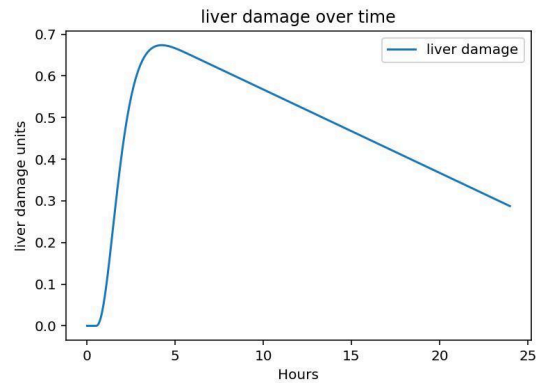
The peak NAPQI level increased from 0.25 mg to 0.7 mg when drinking 305 mmol of ethanol instead of 0. The result of drinking 1830 mmol of ethanol does not significantly differ from drinking 305 mmol of ethanol however.

Is there a safe level of alcohol consumption when taking acetaminophen, or is any consumption a concern?

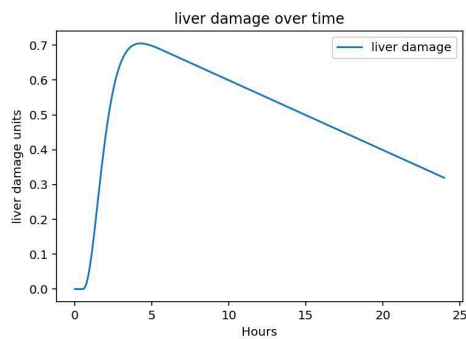
0 mmol of ethanol



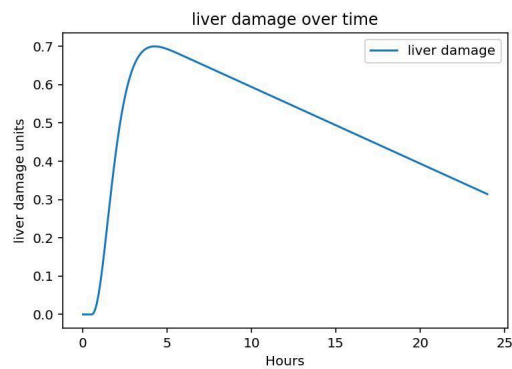
305 mmol of ethanol



1830 mmol of ethanol



2500 mmol of ethanol

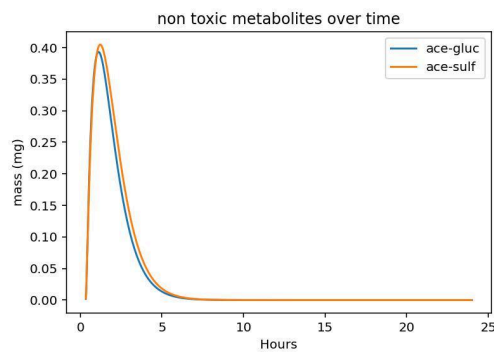
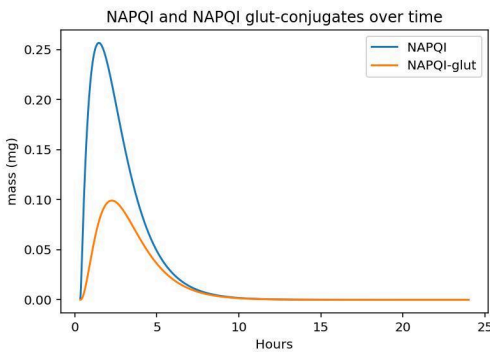


It seems that any amount of alcohol present in the system extends the tail of the liver damage graph, along with increasing the amount of damage ten fold, demonstrating that more damage is caused over-time and any consumption of alcohol is unsafe.

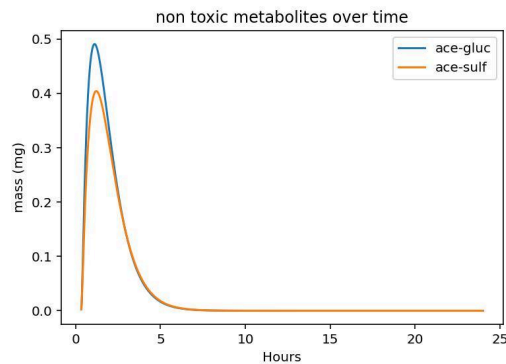
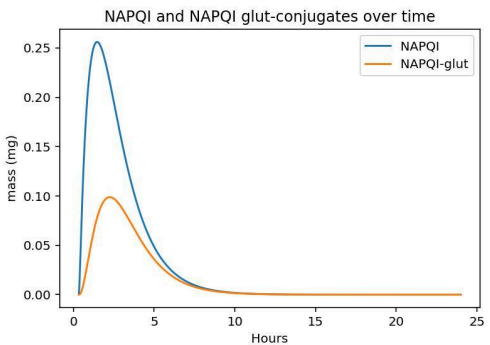
How does greater availability of metabolization pathways affect the metabolization process?

The three metabolization pathways in this simulation are Glucuronidation, Sulfation, and Oxidation (P450). We expect greater availability of metabolization pathways to allow the subject to metabolize acetaminophen more effectively and have lower risk of liver damage, except for the oxidation pathway since that is the one generating NAPQI. The following are the result of control group simulation with no changes, `glu_max_met_rate` increased by 25%, `sulf_max_met_rate` multiplied by 25%, and `p450_max_met_rate` multiplied by 25%:

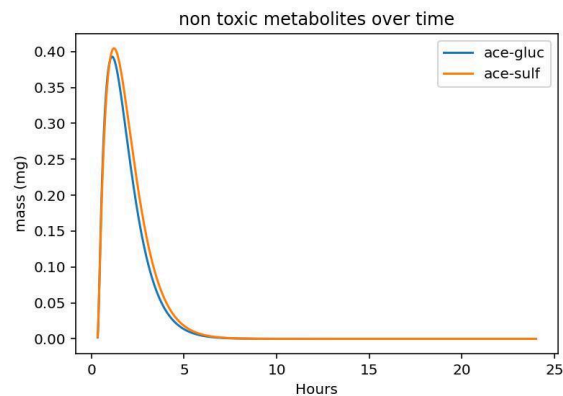
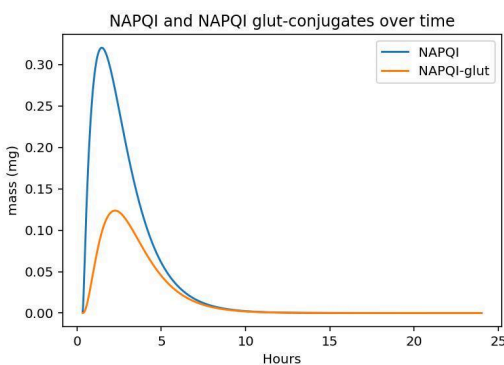
Control (1 dose of 650 mg):



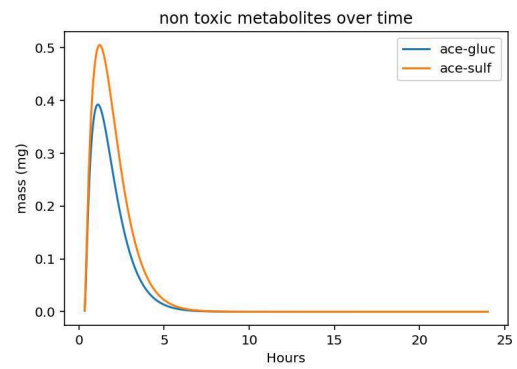
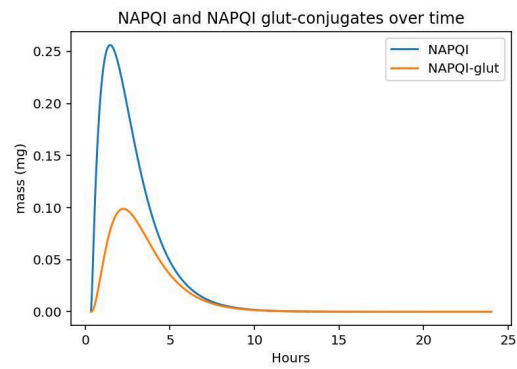
Gluc 25%:



P450 25%:



Sulf 25%:

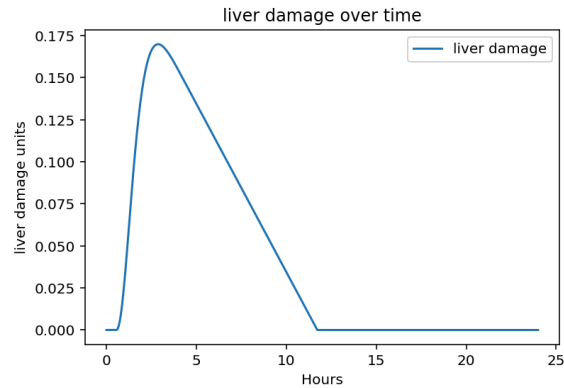


Increasing the max metabolism rate for each pathway causes a higher peak for the corresponding metabolites as expected.

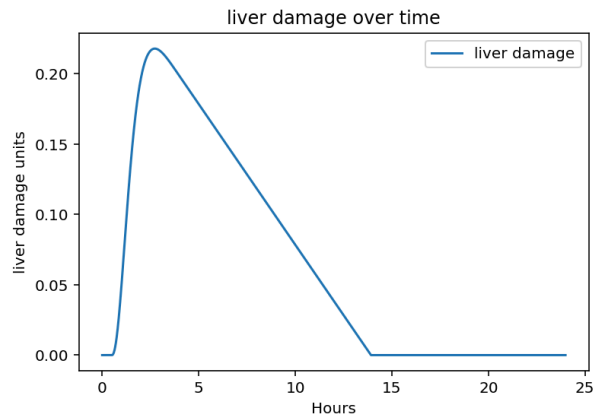
How do multiple factors (age, weight, gender) affect metabolism?

Adjusting variables to meet these changes, namely liver size and blood volume to match these cases (children and the elderly have smaller livers, lower weight has a smaller liver, females have a smaller liver), we found that any of the cases that cause a smaller liver causes more damage over time from the same dose.

1.0 L Liver volume



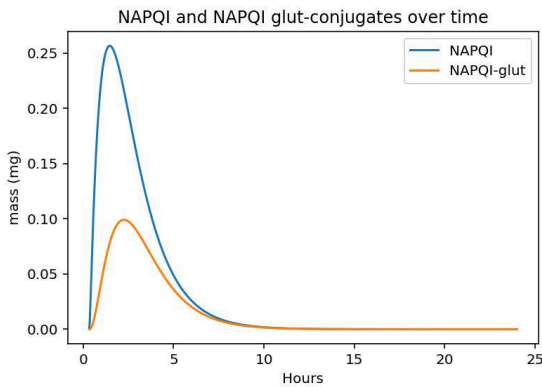
0.8 L Liver volume



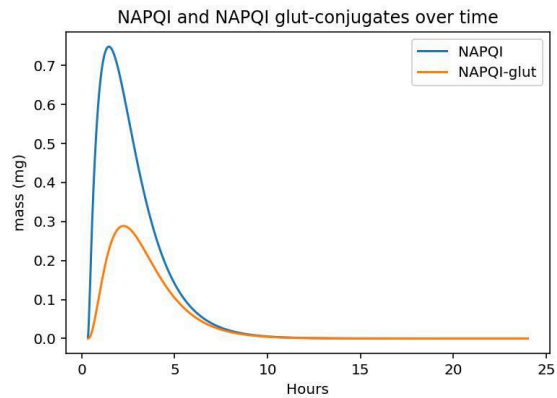
How much faster does NAPQI accumulate when CYP2E1 is more expressed?

CYP2E1 expression is increased during the consumption of alcohol, so consumption of alcohol will increase the processing of acetaminophen into NAPQI.

0 mmol of ethanol



305 mmol of ethanol

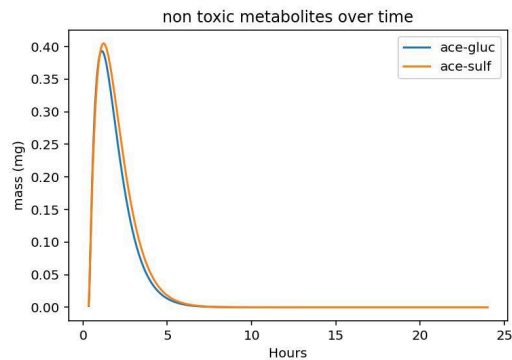
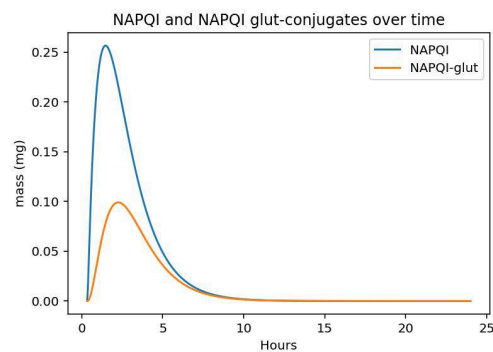


We see this during alcohol consumption, where CYP2E1 is more expressed. NAPQI rises almost three times as much as compared to baseline.

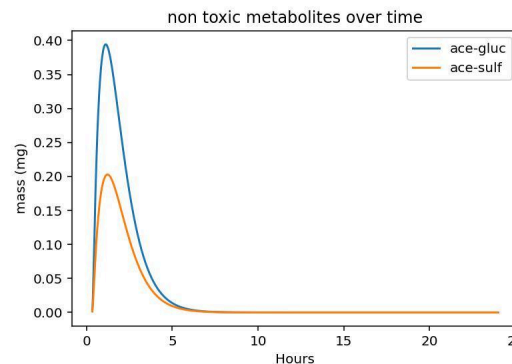
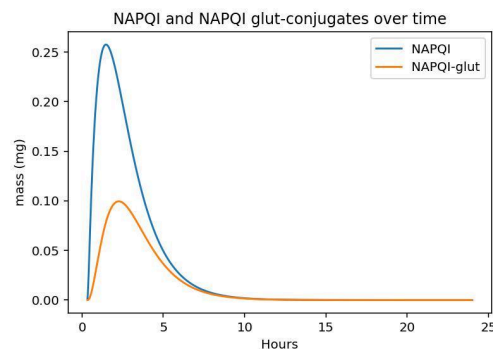
How does a lesser sulfation capacity affect the ability of the body to process acetaminophen and the levels of NAPQI?

Lower sulfation capacity would result in faster saturation of that pathway, which means more Glucuronidation and Oxidation, so We expect there be an increase in NAPQI as well as ace_gluc. This is the result of control group simulation with no modification and sulfation max metabolism rate decreased to 50%:

Control (1 dose of 650 mg):



Sulf decreased by 50%:

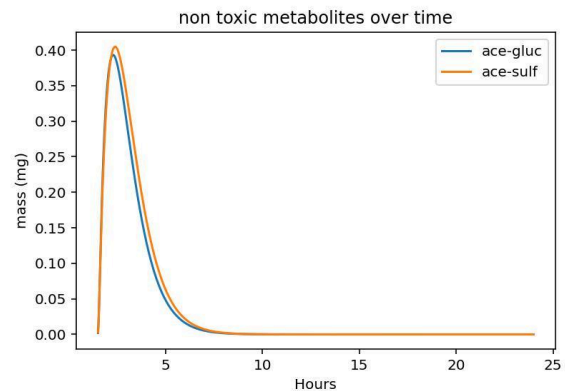
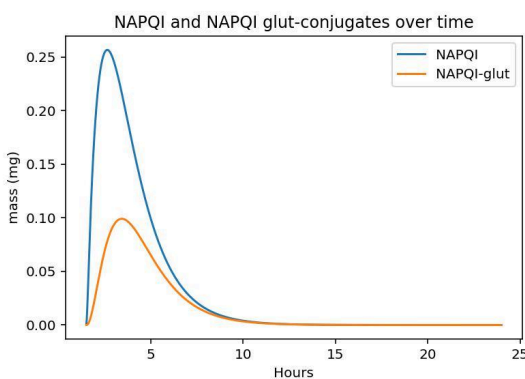


It turns out that when taking a normal dose of acetaminophen, there is no significant difference in NAPQI which was generated.

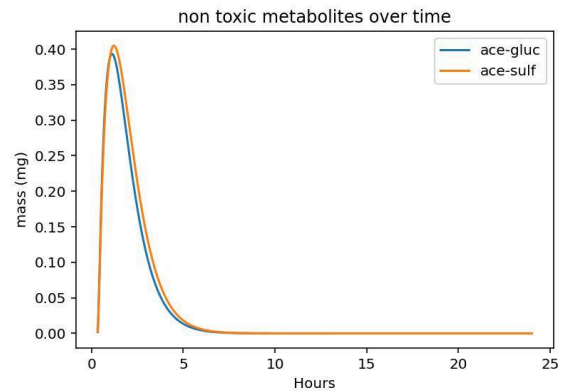
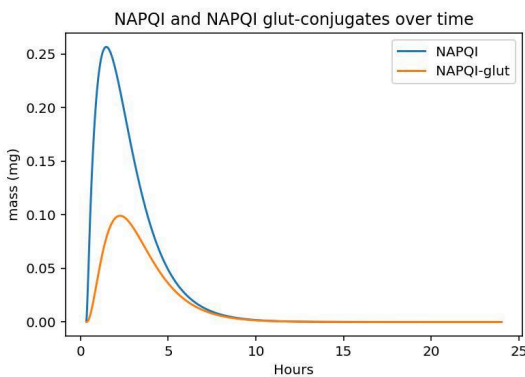
How does digestion affect acetaminophen metabolism

The absorption of acetaminophen in the intestines can take between 20 to 90 mins based on if the subject had been fed or not. Based on this information, the metabolization of acetaminophen should just occur about an hour later if the subject was fed instead of not being fed. The following are the result of being fed or not being fed along with consuming 650 mg of acetaminophen once.

Metabolites generated after being fed



Metabolites generated after not being fed

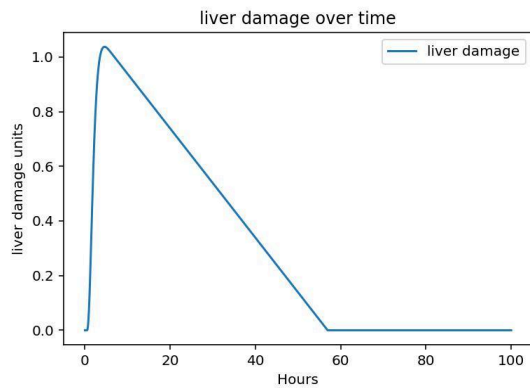


The results of the simulations are nearly identical, but the metabolites start appearing a bit later when being fed.

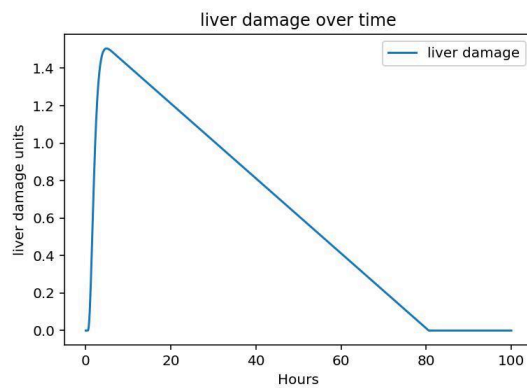
What is the maximum amount of acetaminophen you can take without lasting liver damage?

Our simulation is not able to simulate a situation where the damage is permanent, and all damage is eventually healed as long as the simulation time allows for it. Liver damage above 1 begins around a dose of 2500 mg. According to Medilne Plus, Adults should not take more than 3000 mg of acetaminophen a day. (<https://medlineplus.gov/ency/article/002598.htm>) the damage reaches about 1.5 units

Liver damage at 2500 mg

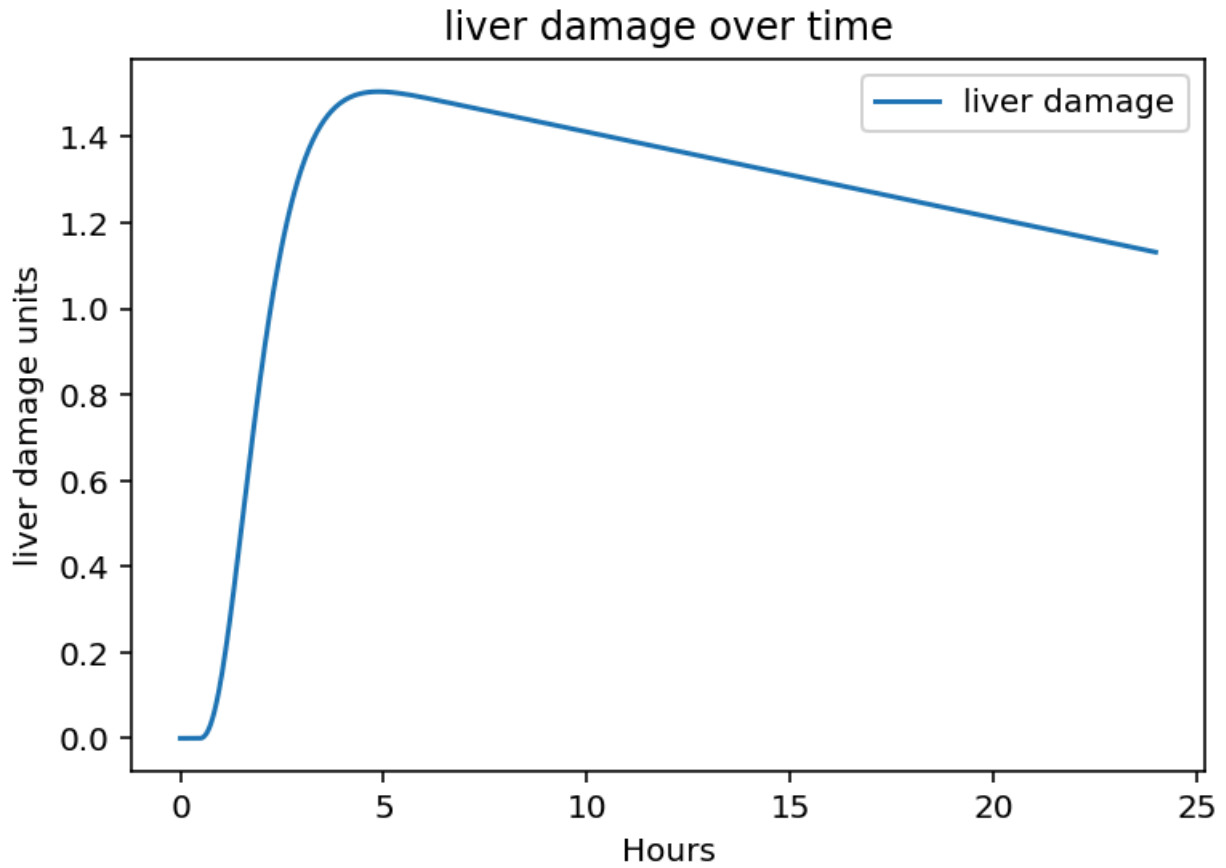


Liver damage at 3000 mg



How fast does damage occur in overdose?

Damage seems to start occurring about one hour after the dose was taken, and peaks at 5 hours after the dose was taken



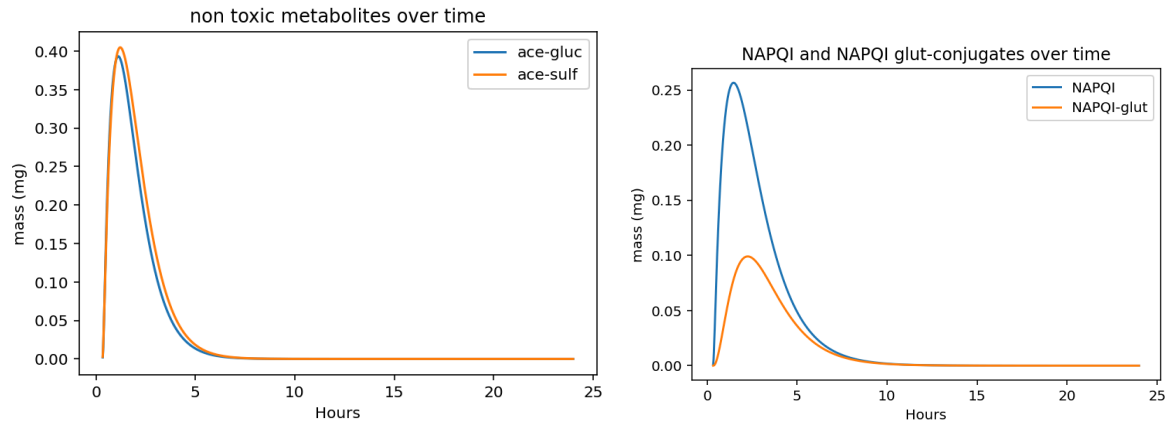
Is the recommended dose of acetaminophen safe or could it be dangerous in some cases?

Through the course of our testing, we found several cases where someone taking the recommended dose could be dangerous. The most prominent one was if they had consumed alcohol recently, as that makes the liver produce more NAPQI and eliminate it less. Another situation was if they have a smaller liver, such as children. This leads to less ability to metabolize which allows NAPQI to sit unconjugated for longer.

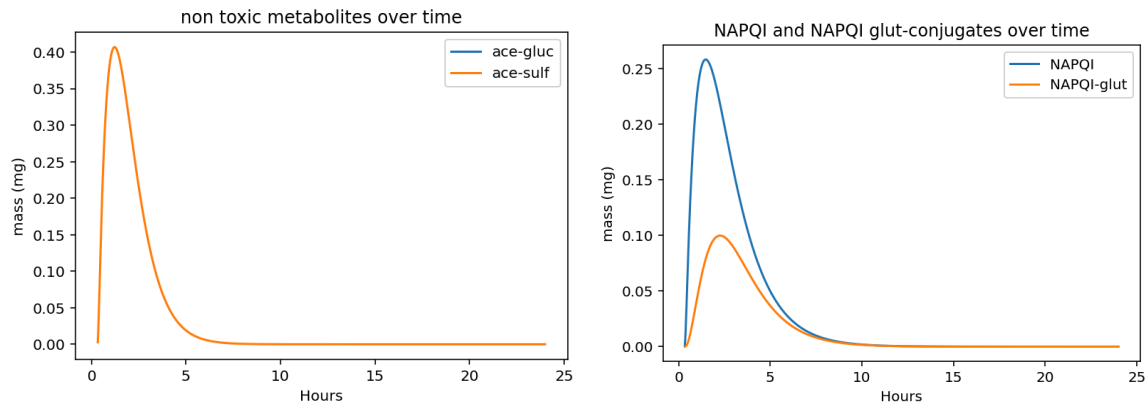
What happens when glucuronidation is saturated?

In order to test this, we used a control run of a single 650 mg dose and a run of the simulation with `glu_max_met_rate` set to 0, to simulate the pathway being saturated and thus not able to handle any more acetaminophen. The results are as follows:

Non-modified



Modified:

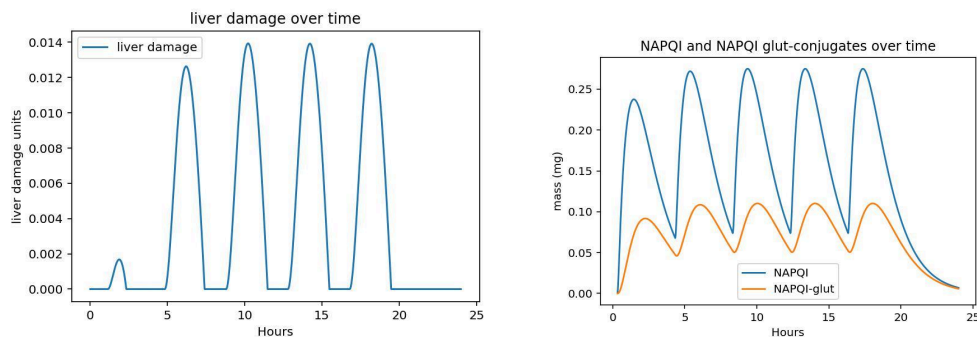


Interestingly, the peak amount of NAPQI or acetaminophen-sulfation conjugate do not increase, though acetaminophen-glucuronidation conjugates are not produced as the pathway is "saturated".

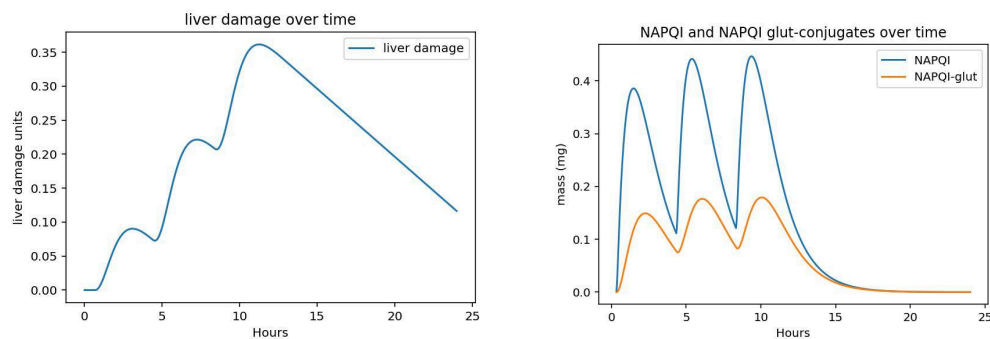
Is one large dose safer or more dangerous compared to multiple smaller doses?

Adults should not take more than 3000 mg of acetaminophen a day. But there may be difference between taking 3000 once a day, and taking 1000 mg 3 times a day. We hypothesize that taking smaller doses more frequently is safer than taking large doses less frequently. The following are the result of liver damage when taking 600 mg 5 times a day, 1000 mg 3 times a day, and 3000 mg once a day:

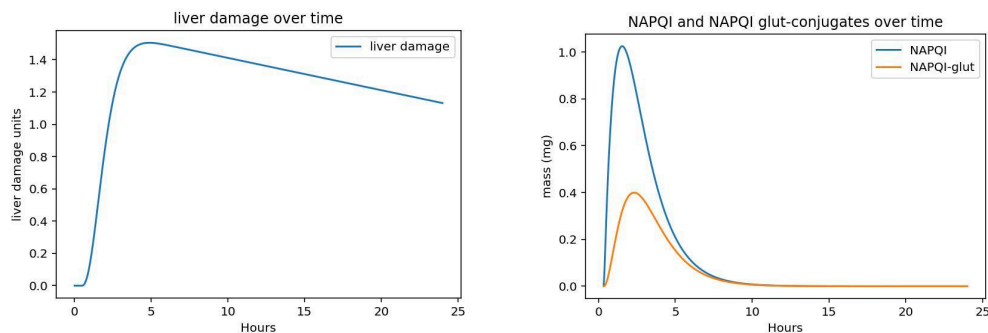
600 mg 5 times a day:



1000 mg 3 times a day:



3000 mg once a day:



The peak NAPQI and damage are highest when taking one large dose is much higher than when taking smaller doses.

Activity History & Repository

Repository: <https://github.com/LeeKaiser/AcetaminophenProject>

Commit history:

Kaiser was working on blood stream, non toxic metabolization pathways in liver, main, plotting, and plot saving. Vivian was working on digestive system, NAPQI and liver damage in liver, Kidneys, and some of the plotting.

The commit history screenshots are listed below:



Commits on Dec 3, 2025

changed digestive system to use LeeKaiser committed 2 hours ago	e124c1f		
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Commits on Dec 2, 2025

liver damage updates Vivian committed yesterday	2f3a76c		
added graph for liver damage LeeKaiser committed yesterday	70f9a11		
removed test figures LeeKaiser committed yesterday	8cdf2ca		
added custom directory generation LeeKaiser committed yesterday	7593860		
added another plot gen method LeeKaiser committed yesterday	d427e08		
added graph saving WIP LeeKaiser committed yesterday	5e87b4c		
convert to masked array WIP LeeKaiser committed 2 days ago	0cba307		
added some comments LeeKaiser committed 2 days ago	d897f07		

Commits on Dec 1, 2025

kidneys added Vivian committed 2 days ago	5ccf45b		
variables testing Vivian committed 2 days ago	1fb5214		
digestive system plotting Vivian committed 2 days ago	fa5b560		
GSH regen, ethanol interaction, and liver damage measuring Vivian committed 2 days ago	07acb17		
updated simulation WIP LeeKaiser committed 3 days ago	e012d18		

Commits on Nov 30, 2025

updated the plots LeeKaiser committed 3 days ago	3050de1		
Merge pull request #9 from LeeKaiser/kaiser/plots WIP LeeKaiser authored 3 days ago	Verified 06721fd		
added plots to main LeeKaiser committed 3 days ago	c7744c2		
NAPQI and glutathione processing Vivian committed 3 days ago	0bb8e71		

Commits on Nov 29, 2025

Merge pull request #8 from LeeKaiser/kaiser/liver

LeeKaiser authored 4 days ago

Verified

2ffe23e



updated liver to return

LeeKaiser committed 4 days ago

a34f7b3



Merge pull request #7 from LeeKaiser/kaiser/liver

LeeKaiser authored 4 days ago

Verified

f36c5bb



added non toxic metabolism of ace

LeeKaiser committed 4 days ago

e45bbb5



Update .gitignore

LeeKaiser committed 4 days ago

0a3482a



updated main and added comments for blood stream methods

LeeKaiser committed 4 days ago

d2aaec2



unit confusion correction

Vivian committed 4 days ago

0a06495



Merge pull request #6 from LeeKaiser/kaiser/bloodstream

LeeKaiser authored 4 days ago

Verified

ee12b76



Merge branch 'kaiser/bloodstream' of https://github.com/LeeKaiser/AcetaminophenProject into kaiser/bloodstream

LeeKaiser committed 4 days ago

94a6c31



updated blood to use its own ace_in_liver var and now returns change

LeeKaiser committed 4 days ago

f1486f4



Merge pull request #5 from LeeKaiser/kaiser/bloodstream

LeeKaiser authored 4 days ago

Verified

5e7ace3



Merge branch 'main' into kaiser/bloodstream

LeeKaiser authored 4 days ago

Verified

cd8ff2f



revert change to pycache

LeeKaiser committed 4 days ago

801c92b



Revert "blood changes"

LeeKaiser committed 4 days ago

a3ffee6



blood changes

LeeKaiser committed 4 days ago

cf0aea2



Commits on Nov 28, 2025

blood changes

LeeKaiser committed 5 days ago

15982cd



added acetaminophen going from blood to liver

LeeKaiser committed 5 days ago

0a3220e



Commits on Nov 28, 2025

updated main to use updated digestive system correctly

LeeKaiser committed 5 days ago

fdfffbc



corrected a mistake in giveDose

Vivian committed 5 days ago

5c0f6c6



comments and doc

Vivian committed 5 days ago

c949f8d



finished intestine, revamped stomach, cleaned up some

Vivian committed 5 days ago

bfc8da3



stomach digestive system done

Vivian committed 5 days ago

20c5817



decompisition and variables setup

Vivian committed 5 days ago

7315473



Commits on Nov 26, 2025

Merge pull request #4 from LeeKaiser/kaiser/mainchanges

LeeKaiser authored last week

Verified

a477e9e



added step

LeeKaiser committed last week

93bb890



Merge pull request #3 from LeeKaiser/kaiser/mainchanges

LeeKaiser authored last week

Verified

b4402fd



added global vars file

LeeKaiser committed last week

bcf303c



Commits on Nov 25, 2025

Merge pull request #2 from LeeKaiser/kaiser/mainchanges

LeeKaiser authored last week

Verified

cee5046



Update Blood_Stream.py

LeeKaiser committed last week

48b4fea



Commits on Nov 24, 2025

Merge pull request #1 from LeeKaiser/kaiser/mainchanges

LeeKaiser authored last week

Verified

38d0a2a



added initial variables for main

LeeKaiser committed last week

4ef88d9



added starting files to complete first benchmark

LeeKaiser committed last week

eedb01a



Initial commit

LeeKaiser authored last week

Verified

849b076

